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Research on Anomaly Detection and Privacy Protection of Network Security Data Based on Machine Learning

Shuang Yuan*

Technology Risk Management, American Airlines, Fort Worth, Texas, 76155, United States

Abstract

With the increasingly complex network security situation, the traditional protection technology is gradually unable to cope with new and changeable network attacks. The anomaly detection method based on machine learning has become an important tool to improve the network security protection ability. With the wide application of cloud computing, how to give full play to the advantages of machine learning under the premise of protecting data privacy has become an important issue in current research. To solve this challenge, a privacy protection scheme based on homomorphic encryption is proposed to ensure the privacy and security of user data by anomaly detection of encrypted data without affecting the detection accuracy. In order to solve the format mismatch problem of encrypted data, this paper also designs a format verification mechanism to detect and correct format errors before data upload to ensure the accuracy of the training process. The experimental results show that this method can achieve the same detection effect as the traditional method while protecting privacy and has a good application prospect. It not only provides a feasible privacy protection scheme for the field of network security, but also provides a valuable reference for other fields involving sensitive data processing.

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* Corresponding author. Tel.: +0-000-000-0000 ; fax: +0-000-000-0000 .

E-mail address: shawyuan02@gmail.com

1. Introduction

In order to improve network defense capabilities, more and more enterprises and scientific research institutions rely on machine learning technology. With its powerful data processing and pattern recognition capabilities, this technology can effectively detect network intrusion and abnormal behavior. Machine learning can automatically identify potential security threats through learning historical data, thus providing effective solutions to prevent network attacks. However, with the data transmission and storage gradually relying on cloud computing platforms, machine learning can prevent network attacks. How to make full use of cloud computing resources to efficiently complete machine learning tasks under the premise of ensuring data security has become an urgent problem to be solved.

In the cloud computing environment, users upload data to remote servers for processing, resulting in more limited control over data. Although the cloud computing platform provides powerful computing power, data privacy and security issues have become new hidden dangers due to the comprehensive control of data storage and processing by cloud service providers, especially when it comes to sensitive data processing. How to prevent data leakage and abuse has become the core issue of privacy protection. In order to effectively utilize cloud computing resources without exposing the original data, homomorphic encryption technology, as an encryption method, can perform calculations in the state of data encryption, thus effectively protecting data privacy, and has become one of the important technologies in current research.

However, even if encryption technology can guarantee the privacy of data calculation process, there is still a key problem - the rationality verification of encrypted data, in machine learning tasks, uploaded data may contain errors or non-compliant samples, if these problems are not found in time, may have a negative impact on the training results, and even lead to more serious security risks. Existing studies mainly focus on data privacy protection, but there is still a lack of systematic solution for verifying the rationality of encrypted data. Therefore, this paper proposes a machine learning scheme for privacy protection based on homomorphic encryption, combined with data format checking mechanism, to verify the rationality of data while ensuring data encryption.

This paper designs a training scheme of logistic regression and convolutional neural network for privacy protection, encrypts data with partial homologous encryption technology, and designs a ciphertext data format verification mechanism to ensure that the data uploaded to the cloud complies with the predefined specifications. Through experimental verification, the privacy protection method proposed in this paper can not only effectively guarantee data security, but also ensure that the data uploaded to the cloud complies with the predefined specifications. Meanwhile, the data format checking mechanism can effectively prevent the interference of malicious data, thus improving the robustness of the system.

2. Relevant research

In recent years, in the field of network security, anomaly detection technologies for different application scenarios have been continuously developed and innovated. AA Abdallah^[1] proposed an efficient black hole attack identification method by analyzing the behavior characteristics of attackers. This method can accurately detect black hole attacks in mobile AD hoc networks through the identification of node behavior characteristics. At the same time, R Liu's^[2] research explored various applications of machine learning in network security, focusing on the analysis of deep learning, integration methods, and the application of supervised and unsupervised learning in anomaly detection and defense, in-depth discussion of the research challenges and future development directions of these methods in the integration of security frameworks, and provided new perspectives and ideas for network security research.

In the field of Internet of Things (IoT), B Liu^[3] proposed a security tradeoff model based on linear programming to reduce the impact of high false positives on iot data availability, and improve the security and data availability of iot systems through this model. In addition, Y Zhang^[4] proposed a new federated learning method, FedGroup. By updating the central model according to the learning mode of different iot device groups, the accuracy of anomaly detection in smart home is greatly improved.

For the network security of smart cities, I Priyadarshini^[5] proposed a solution for detecting network attacks of Internet of Things devices by combining the methods of federation learning and partition learning. This method not

only improves the detection performance, but also takes into account the problem of data privacy protection, providing a scalable solution that can both improve security and ensure privacy. It further promotes the technological development in the field of smart city Internet of Things security.

In the field of computer network security, S Wang^[6] proposes an innovative fusion model that combines isolated forests, generative adversative networks (Gans), and Transformers to solve high-dimensional data, complex network topologies, and time series data processing problems in computer network anomaly detection and log analysis. The application of unsupervised learning methods in network anomaly detection has also become a research hotspot. N Sharma^[7] conducted a comprehensive analysis on the performance of various unsupervised anomaly detection algorithms on different data sets, and discussed the challenges and potential defects faced by current unsupervised learning methods. The research shows that unsupervised learning has significant effectiveness in the field of network security. It also provides valuable experience and enlightenment for the future research direction.

In addition, P Bretschneider^[8] proposed a system that combines network intrusion and phasor data anomaly detection. The system monitors network packets and measurement information through expert rules, a class of classifiers and recurrent neural networks, which provides strong support for ensuring the safe operation of dynamic control center in the face of network attacks. The real time simulation test shows that the system has good adaptability and effect under various operating and network attack conditions. T Li^[9] proposed a task-oriented abnormal packet behavior detection method. This method, through fine-grained task execution monitoring and embedded security label technology, combined with hash value matching and finite state machine mechanism, can accurately improve packet authentication accuracy and improve the accuracy of abnormal behavior determination, providing an effective guarantee for the safe execution of network tasks.

Finally, S Chakravarthi^[10] proposed an intrusion detection method based on machine learning, which detects nine types of attacks through two stages, combines historical network traffic data and Internet traffic data, and builds an effective network intrusion detection mechanism. This method provides an effective technical solution to solve the security and privacy problems in cloud computing services. To sum up, the above research results have proposed innovative solutions and technologies in anomaly detection in various fields. The application potential of machine learning, deep learning and federated learning technologies in network security has been fully demonstrated. Future research should continue to focus on improving detection accuracy and reducing false positives while paying more attention to privacy protection and data security. To provide more intelligent and efficient solutions to the complex and changing network security challenges.

3. Machine learning scheme for privacy protection based on homomorphic encryption

3.1 Program overview and system design

The design of the system involves several key roles, including key generation center (KGC), data provider (DP), storage server (CSS), encryption service provider (CSP), service forwarding party (SF), and service requester (SR). Among them, KGC is responsible for generating and distributing public and private key pairs and keeping them offline after key distribution is complete, while DP encrypts the data using the public key before storing it, ensuring that the data remains private during transmission. The encrypted data is stored in the CSS, which provides data storage and computation services in conjunction with the CSP, who assists with model training while performing the computation tasks. The SF is responsible for receiving the encrypted model for inference service, and finally forward the service request to the SR, and the SR provides the data and performs the classification operation, through this complete process to ensure the security and efficiency of data processing.

In the design of system roles, all participants can be divided into two types of trusted and semi-honest roles, KGC is considered to be completely trusted because its generated keys do not reveal any information, while DP, CSS, CSP, SF, and SR are considered to be semi-honest roles, although they follow the protocol to perform tasks. But there may be an attempt to get more information from the data stored or the services provided. With this in mind, the system assumes that there is no malicious collusion between CSP and CSS, that an attacker may steal DP plaintext data from stored encrypted data, or attack SF to obtain prediction results, that an attacker may also try to eavesdrop on the data transmission process, or attack CSS to obtain stored ciphertext data and its plaintext correspondence. In addition,

the system assumes that the attacker does not control both CSS and CSP to prevent more sophisticated attacks from occurring.

In order to protect data privacy, the system adopts partial homomorphic encryption technology, which enables it to perform privacy-protecting machine learning tasks in the state of encrypted data. In the initialization stage of the system, KGC generates and allocates public and private key pairs for each participant, and each participant uses these keys to encrypt their data. In the calculation process, some homomorphic encryption can support addition and subtraction operations on encrypted data, but does not support multiplication or power operations. Therefore, the system designs a series of encryption calculation protocols. Including secure integer multiplication, vector inner product calculation, integer comparison, and sigmoid function, these protocols ensure that all parties can perform the necessary computation operations without exposing data privacy while maintaining the encrypted state of the data. When dealing with floating point numbers, the system converts them to integer form, and reduces the precision loss in the conversion process by setting the precision value, so as to ensure that the whole calculation process does not affect the accuracy and validity of the data while protecting the privacy.

Through these encryption mechanisms, the system effectively guarantees the security of data privacy in the process of data storage, transmission and calculation, and supports the training and prediction tasks of machine learning algorithms, ensuring that efficient calculation and analysis can be carried out under the premise of privacy protection, so as to provide users with safe and reliable services.

3.2 Security and Performance Analysis

Through the security verification of several protocols, especially the secure integer protocol (SIM) and the secure vector inner product protocol (SVIP), the analysis is carried out using the method of comparing the ideal world with the real world. Assuming that the implementors of the protocol are CSS and CSP, the security of the protocol is proved by constructing a simulable adversary model in polynomial time. In the privacy analysis, theorem 1 shows that when DTPKC is semantically secure, the SIM protocol can safely perform ciphertext multiplication under the semi-honest adversary model. In this process, the simulator ensures the security of the protocol by initializing the necessary parameters, verifying the received message, and partially decrypting the message if it is valid.

In analyzing the security of the PPLR protocol, we consider a scenario where the adversary is capable of intercepting communication but lacks access to the private key or decryption credentials of the SF, which ensures the protection of the training data's confidentiality. Even if the encrypted data is intercepted, the attacker is unable to retrieve any meaningful information due to the absence of the necessary decryption key, and the robust semantic security provided by DTPKC guarantees effective data protection. In the case of the PPNN protocol, the image data is encrypted through DTPKC, ensuring that neither the CSS nor the CSP has any access to the raw data or model parameters, and the protocol's security module is responsible for safeguarding the computations of all layers, including convolutional, activation, pooling, and fully connected layers, which guarantees complete protection for the privacy of data throughout the neural network training process.

To comprehensively evaluate the performance of the scheme, we employ two neural network models, CNN1 and CNN2, in the experiment, where CNN1 is a basic convolutional neural network comprising convolutional, pooling, activation, and fully connected layers, and CNN2 enhances CNN1 by adding additional convolutional layers and increasing the number of convolutional filters, thus improving the network's depth and capacity. Furthermore, the running time of the DTPFCC algorithm shows a gradual increase as the parameter N is scaled up. As shown in Figure 1, CNN2 consists of ten layers, including two convolutional layers, two pooling layers, three activation layers, and two fully connected layers, with the convolutional layers utilizing nine convolutional filters, and the final fully connected layer consisting of 128 nodes.

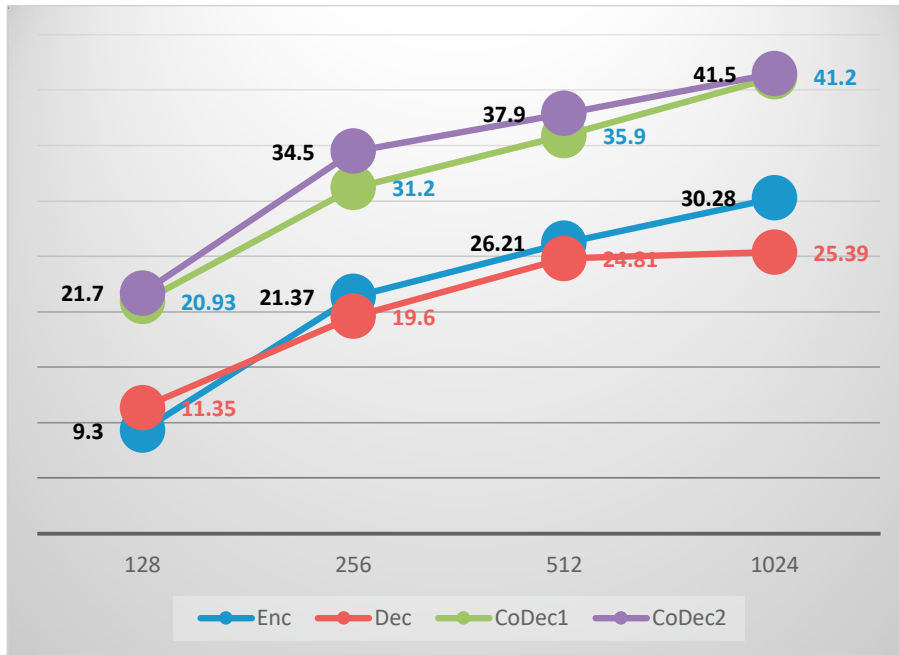


Figure 1. Change of running time of DTPFCC algorithm

Table 1. Calculation cost of SIM protocol (seconds)

Dimension	N=128	N=256	N=512	N=1024
2	0.04	1.10	8.30	68.39
3	0.06	1.17	9.60	70.16
4	0.10	1.23	9.89	74.16
8	0.14	1.39	10.89	80.23
16	0.32	2.75	11.37	158.36

Table 2. Calculation cost of SVIP protocol (seconds)

Dimension	N=128	N=256	N=512	N=1024
2	0.07	1.11	8.13	66.62
3	0.10	1.18	8.87	62.62
4	0.14	1.24	9.25	93.93
8	0.21	1.84	11.36	73.92
16	0.67	2.86	18.36	177.55

Table 3. Calculation cost of the two protocols (seconds)

Dimension	Agreement	N=128	N=256	N=512	N=1024
2	Sigmoid	0.07	1.71	8.22	64.38
2	Scomp	0.31	1.02	9.56	65.44
3	Sigmoid	0.04	1.22	9.56	129.91
3	Scomp	0.45	1.27	10.51	72.38
4	Sigmoid	0.37	1.53	9.63	71.77
4	Scomp	0.58	1.82	11.67	92.56
8	Sigmoid	0.83	2.11	10.63	180.69
8	Scomp	0.76	2.41	11.34	186.24
16	Sigmoid	1.28	2.75	18.19	257.70
16	Scomp	1.28	3.21	25.17	248.83

For secure computing protocols, this study focuses on evaluating security during data processing to influence the performance of secure LR and CNN models. The calculation cost of different protocols was tested in the experiment, as shown in Table 1, 2 and 3. The impact of data dimension and bits of protocol parameter N on the execution time was mainly investigated. The test results show that the computation time increases significantly with the increase of N digits. For example, when N is 128 bits, the execution time is shorter, but when N is increased to 1024 bits, the execution time increases significantly, indicating that the computing overhead of the protocol is closely related to the length of N. The dimension of data set also affects the calculation cost. When testing on MNIST and CIFAR-10 data sets, the calculation time is significantly increased compared with MNIST due to the large amount of CIFAR-10 data. The running time of CNN1 and CNN2 on MNIST is 34.28 seconds and 98.56 seconds, respectively.

We tested the MNIST and CIFAR-10 datasets to evaluate the predictive run time of two types of convolutional neural networks (CNN-1 and CNN-2). When the sample size is 256, the prediction time of CNN-1 and CNN-2 on the MNIST data set is 34.28 seconds and 98.56 seconds respectively, while the prediction time of CNN-2 on the CIFAR-10 data set is 157.29 seconds and 266.38 seconds respectively. The input image size of the CIFAR-10 dataset is $32 \times 32 \times 3$ and the run time is longer than MNIST's $28 \times 28 \times 1$. Through further experiments, we evaluated the performance of privacy protection logistic regression (PPLR) algorithm in Windows 10 operating system, and the results showed that the classification accuracy gradually improved with the increase of iterations, but the accuracy slightly decreased compared with the traditional logistic regression algorithm. This is due to the calculation error caused by using the Taylor approximation to deal with the nonlinear function.

Table 4. Training time consumption of different N bits in different data sets (unit: millisecond)

Data set	Bits of N	PPLR (ms)	PPSVM (ms)	PPLL (ms)	SVM (ms)	LR (ms)	LL (ms)
Iris	128	2438	2705	2566	1358	1021	1129
	256	13383	15904	14072			
	512	111032	127556	118351			
Heart	128	2930	3028	2799	1826	1537	1493
	256	17441	19566	16582			
	512	192320	199558	202596			
Horse Colic	128	3790	4024	3562	2367	2184	2259
	256	25964	29568	24485			
	512	249856	266551	245360			

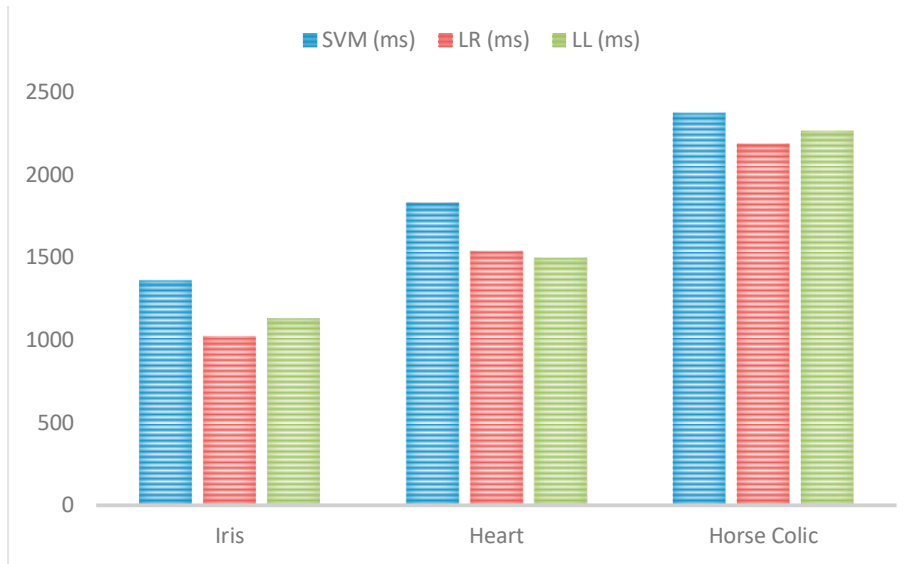


Figure 2. Algorithm training time consumption of different data sets (unit: millisecond)

In addition, the training time of the PPLR algorithm is significantly higher than that of the traditional logistic regression algorithm, and the training time increases with the increase of the number of bits of the safety parameter N . However, under the premise of privacy protection, PPLR algorithm can still maintain good classification performance. Through testing on multiple datasets, such as Iris, Heart, MNIST, and Horse Colic, as shown in Table 4 and Figure 2, the training time of the privacy-protecting algorithm increases with the amount of data and bit length, but remains effective at the task within an acceptable time frame. Although privacy-protecting machine learning algorithms have a high demand for computing resources, they can protect data privacy while maintaining a good classification effect.

4. Training Data Format Check and Privacy Protection Training System Based On Homomorphic Encryption

4.1 Training data format check scheme and implementation

This study proposes a novel solution to optimize data format validation in machine learning training, especially for discrete and continuous data processing problems. The scheme combines homomorphic encryption and short signature technology to ensure the privacy protection of data in the training process, and also effectively verifies the compliance of data format. Short signature technology simplifies the verification of discrete data formats and reduces the risk of format errors. Homomorphic encryption can effectively protect continuous data from being leaked during the authentication process. The experimental results show that this scheme not only improves the accuracy of data format verification but also verifies its feasibility in practical applications.

This solution uses a multi-level system architecture to achieve this goal, including a key generation center, data provider, data validator, auxiliary validator, and cloud platform. The key generation center is responsible for managing and distributing keys and providing training data to data providers. The data validator and auxiliary validator jointly undertake the task of data format validation to ensure the validity and security of the data. Once the data is validated, the cloud platform will use it for further machine learning model training. The combination of homomorphic encryption technology and short signature technology ensures the privacy of data exchange and avoids the risk of data tampering or leakage during transmission.

In order to evaluate the actual performance of the scheme, the algorithm was validated through numerical simulation experiments. The computer is equipped with a 3.20GHz processor and 16GB of memory to ensure the

smooth progress of the experiment. The selected experimental dataset is the Iris dataset, which is widely used in the field of machine learning and contains 150 samples covering three types of iris flowers. Each sample includes four characteristics: calyx length, calyx width, petal length and petal width. The purpose of the experiment is to investigate the effect of illegal data on model training, where illegal data refers to data with an input format or type that does not meet the requirements, such as incorrectly entered text in a number field. To this end, three experimental situations are designed: first, all the fields of the record are illegal, second, some fields are randomly selected as illegal data, and third, the illegal data is randomly distributed in different fields. The experimental results show that the amount of illegal data in the data set is negatively correlated with the training accuracy of the model, and the more illegal data, the lower the accuracy of the training results, which emphasizes the importance of format verification and cleaning of the data.

In terms of time efficiency, this study conducted detailed time analysis of the two validation algorithms, and the same data set as above was used during the experiment to ensure the consistency and comparability of the results. The analysis shows that there is a linear relationship between the execution time of discrete verification algorithm and the data amount, while the execution time of continuous verification algorithm is affected by the data bit length and the encryption process. The experimental results further show that the processing time of discrete verification algorithm increases linearly with the increase of data volume, while the time cost of continuous verification algorithm also increases with the increase of bit length. Through comparative analysis, it is found that whether the data is small or large, the proposed verification scheme shows high efficiency in processing. The final results prove the necessity of data format verification in ensuring the accuracy of the training model, and also show that in practical applications, even in the face of large-scale data sets, the proposed verification scheme can still run efficiently and has good practicability and feasibility.

4.2 Design and implementation of privacy protection training system

Through the integration of several functional modules, the system is designed to provide users with a safe and efficient machine learning task execution environment, while ensuring data privacy protection. First, users must verify their identity through registration and login before they can access the system and start training tasks. After login, the system provides data encryption and decryption functions for users to ensure that the uploaded data is encrypted during transmission to prevent leakage. After data is uploaded, the system verifies its format, and only the encrypted data that meets the specified format can enter the subsequent processing process. The verified data is then sent to the cloud server for privacy training, and the system distributes the task among multiple collaborating servers by selecting the appropriate algorithm to ensure that the user's original data remains encrypted throughout the calculation process. After the training task is completed, the system returns the result to the user in encrypted form, and the user can view the result after decrypting it using the private key. The data transmission module ensures the security of the upload and download process, and provides real-time feedback on the upload progress to ensure the smooth progress of the task. On the whole, the system improves the efficiency of task execution while protecting data privacy through the cooperation of various functional modules, providing a safe and convenient solution.

When users access the system for the first time, they need to complete the registration process, first selecting the appropriate identity based on their personal role, and filling in the user name, password and other necessary information, and finally submit a registration request. After the user enters the correct user name and password, the system verifies the user name and password on the client before submitting the file. Then, the system verifies the user name and password on the server. If the authentication fails, the system displays a clear error message to help the user solve the problem. In addition, the system also supports third-party accounts such as QQ and Weibo for quick login, simplifying the login process for users.

In order to ensure data security, the system provides encryption and decryption functions. After users upload data, the system encrypts the data to ensure that the privacy of the data is not disclosed during transmission. During the encryption process, the system displays the message "Encrypting". After the encryption is complete, you can choose whether to decrypt the data. After clicking the decryption button, the system prompts you whether to confirm the decryption to view the calculation result.

In terms of data transmission, the user will upload the encrypted data to the system, the server will receive the corresponding processing and return the result, when the user clicks the download button, the system will transfer

the processed data to the client in encrypted form to ensure data security. During the upload and download process, the system displays the progress to ensure that users can clearly understand the status of data transfer.

In addition, the system provides an online verification function to ensure that the uploaded data is effectively verified before the privacy calculation. After the user uploads the data, the system will automatically verify the accuracy and integrity of the data. After the verification is passed, the user can continue to perform the privacy calculation. If the verification fails, the system deletes the uploaded file and prompts you to upload the required data set again. You can perform subsequent operations only after the verification is successful.

In the privacy computing module, after the user selects the required training algorithm, the system will display the detailed information of the task, including the task progress and status update, and the user can cancel the task at any time, stop the calculation and clear the relevant data. If the calculation result does not meet the user's requirements, the user can re-submit the task. The system updates the task in real time according to the task progress and status, ensuring that the user is always aware of the calculation status.

The system also supports logistic regression and neural network schemes for privacy protection. After the user selects the algorithm, the system will perform prediction calculation based on the encrypted data, and return the predicted results. The user can download and decrypt the calculation results to ensure that the privacy of the data is protected and the accuracy of the calculation results is guaranteed. When processing image data, the system can also analyze and predict under the premise of ensuring privacy, and meet users' processing needs for different types of data.

5. Conclusion and prospect

With the rapid development of cloud computing, machine learning, advanced data analysis capabilities and model training, learning skills have improved significantly. In this article, we introduce an encryption-based learning machine that ensures privacy during training by handling encrypted data securely. The method includes a secure protocol designed to facilitate the encryption process of training models such as logistic regression, neural network carriers while maintaining the confidentiality of relevant data. Experimental results show that this method not only guarantees data security, but also maintains the accuracy of the training process. By validating encrypted data, we can effectively solve problems to ensure the accuracy of data format training and the reliability of training results. This research, development and testing of a data protection training system can effectively process data and obtain accurate results while maintaining privacy. Despite these positive achievements, there is still room for improvement in the handling of non-linear functions, in improving the efficiency of encryption algorithms and in improving the speed of data format validation. Future research should focus on developing more efficient nonlinear functional processing techniques to improve the efficiency of encryption algorithms, speed up the data format authentication process, and improve overall system performance. With the development of AI technology, machine learning systems providing privacy protection are expected to be used in an increasing number of situations. In the future, it will be possible to focus on improving the flexibility and adaptability of the system to the needs of different departments and users, while promoting the continuous development of smart technologies.

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